P275.

5.2

对偶关系：

电偶极子	磁偶极子
$\bar{E}$	$\bar{H}$
$\bar{H}$	$-\bar{E}$
$\bar{J}$	$\bar{M}$
$\bar{A}$	$\bar{F}$
ϵ	μ
μ	ϵ
k	k
η	$\frac{1}{\eta}$
$\frac{1}{\eta}$	η

4.8a–4.10c

$H_r = H_\theta = 0$

$H_\phi = j\frac{kI_0 l\sin\theta}{4\pi r}\left[1+\frac{1}{jkr}\right]e^{-jkr}$

$\bar{E} = \bar{E}_A = \frac{1}{j\omega\epsilon}\nabla\times\bar{H}$

$E_r = \eta\frac{I_0 l\cos\theta}{2\pi r^2}\left[1+\frac{1}{jkr}\right]e^{-jkr}$

$E_\theta = j\eta\frac{kI_0 l\sin\theta}{4\pi r}\left[1+\frac{1}{jkr}-\frac{1}{(kr)^2}\right]e^{-jkr}$

$E_\phi = 0$

对偶 $\Longrightarrow$

$E_r = E_\theta = 0$

$E_\phi = -j\frac{kI_m l\sin\theta}{4\pi r}\left[1+\frac{1}{jkr}\right]e^{-jkr}$

$H_r = \frac{1}{\eta}\frac{I_m l\cos\theta}{2\pi r^2}\left[1+\frac{1}{jkr}\right]e^{-jkr}$

$H_\theta = j\frac{1}{\eta}\frac{kI_m l\sin\theta}{4\pi r}\left[1+\frac{1}{jkr}-\frac{1}{(kr)^2}\right]e^{-jkr}$

$H_\phi = 0$.

与式5-20a～5-20d相同……

5.3 (a) $R_r = 20\pi^2\left(\frac{C}{\lambda}\right)^4$，$C = 2\pi a$. 则 $R_r = 20\pi^2\left(\frac{2\pi\cdot\frac{\lambda}{30}}{\lambda}\right)^4 = 20\pi^2\left(\frac{2\pi}{30}\right)^4$

$= 0.3798\,\Omega$

(b) $R_L = R_{hf}\frac{C}{P}\sqrt{\frac{\omega\mu_0}{2\sigma}} = \frac{2\pi a}{2\pi b}\sqrt{\frac{2\pi f\mu_0}{2\sigma}}$

$= \frac{a}{b}\sqrt{\frac{\pi f\mu_0}{\sigma}} = \frac{\frac{\lambda}{30}}{\frac{\lambda}{1000}}\cdot\sqrt{\frac{\pi\cdot 10\times10^6\times4\pi\times10^{-7}}{5.7\times10^7}} = 0.02774\,\Omega$

(c) $R_m = R_r + R_L = 0.3798\,\Omega + 0.02774\,\Omega = 0.408\,\Omega$.

(d) $X_A = \omega L_A = 2\pi f L_A = 2\pi f\{\mu_0 a[\ln(8\frac{a}{b})-2]\} = 283.139$

CS 扫描全能王
3亿人都在用的扫描App

$$X_i = \omega L_i = \omega\left[\frac{a}{\omega b}\sqrt{\frac{\omega\mu_0}{2\sigma}}\right] = \frac{a}{b}\cdot\sqrt{\frac{\pi f\mu_0}{\sigma}} = \frac{\frac{\lambda}{30}}{\frac{\lambda}{1000}}\cdot\sqrt{\frac{\pi\cdot 10^7\cdot 4\pi\times10^{-7}}{5.7\times10^7}}$$

$$= 0.02774\,\Omega$$

$$X_T = X_A + X_i = 283.1667\,\Omega$$

则 $Z_{in} = (R_L + R_r) + j(X_A + X_i)$

$$= (0.40754 + j\,283.1667)\,\Omega.$$

(e) $e_{cd} = \dfrac{R_r}{R_r + R_L} = \dfrac{0.3798}{0.40754}\times100\% = 93.19\%$

5.7 单匝: $R_r = 20\pi^2\left(\dfrac{C}{\lambda}\right)^4 = 0.3158\,\Omega$

$$R_L = R_{hf} = \frac{a}{b}\sqrt{\frac{\omega\mu_0}{2\sigma}} = \frac{1}{10\pi\times10^{-3}}\times\sqrt{\frac{2\pi\times10^7\times4\pi\times10^{-7}}{2\times5.7\times10^7}}$$

$$= 0.0265\,\Omega.$$

则 $e_{cd} = \dfrac{R_r}{R_r + R_L}\times100\% = 92.26\%$

四匝: $R_r = N^2 R_r = 16\times0.3158\,\Omega = 5.0532\,\Omega.$

$$R_L = \frac{Na}{b}R_s\left(\frac{R_p}{R_0}+1\right)$$

$$R_s = \sqrt{\frac{\omega\mu_0}{2\sigma}} = 8.3223\times10^{-4}\,\Omega,\quad R_0 = \frac{NR_s}{2\pi b} = \frac{4\times(8.3223\times10^{-4})}{2\pi\cdot10^{-3}} = 0.5298\,\Omega$$

$$R_p = 0.5R_0 = 0.2649\,\Omega.$$

则 $R_L = 0.1572\,\Omega.$

则 $e_{cd} = \dfrac{R_r}{R_r + R_L}\times100\% = \dfrac{5.0532}{5.0532 + 0.1572}\times100\% = 96.98\%$

CS 扫描全能王
3亿人都在用的扫描App